COMPLETE DATA ANALYSIS PORTFOLIO

Five domains. Five evidence-based analyses.

A reproducible portfolio exploring historical retail transactions, hospital encounters, NBA outcomes, equity prices, and online shopping sessions.

BUSINESS SALES

GBP gross line sales: 20,972,594.57 GBP

2009-12-01; 2011-12-09; invoice line

HEALTHCARE

30-day readmission rate: 11.3%

1999; 2008; eligible hospital encounter; discharge disposition ID != 11

SPORTS

Historical Elo holdout accuracy: 66.4%

1947; 2015; 1977; 2022; team-game and player-season (separate)

FINANCE

Illustrative annualized price return: 14.4%

2010-01-04; 2016-12-30; security-date; split-adjusted price return; dividend policy undocumented

E-COMMERCE

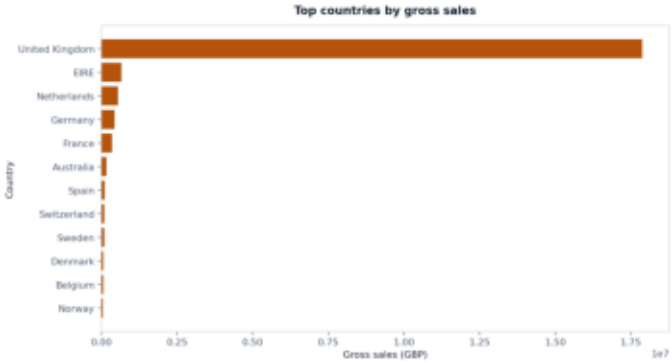
Session conversion rate: 15.5%

session; one-year source sample

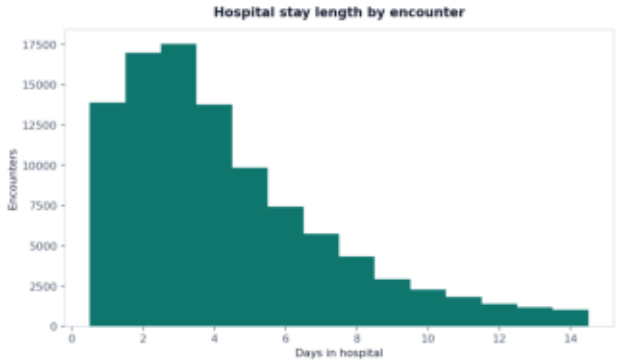
Interpretation

Results are source-specific and historical. Denominators, units, source coverage, uncertainty, and limitations differ by domain. No combined cross-domain score is meaningful. Revenue is not profit; hospital associations are not clinical guidance; historical returns are not investment advice; e-commerce sessions do not measure repeat buying.

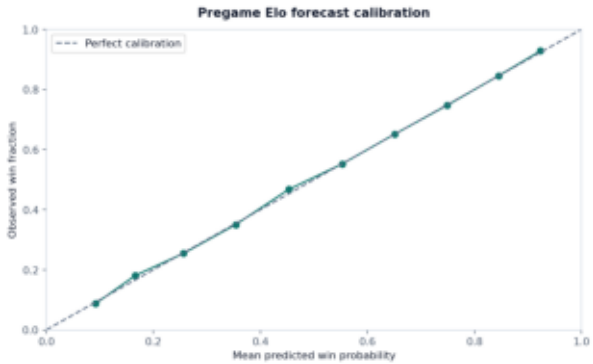
SELECTED EVIDENCE



BUSINESS SALES



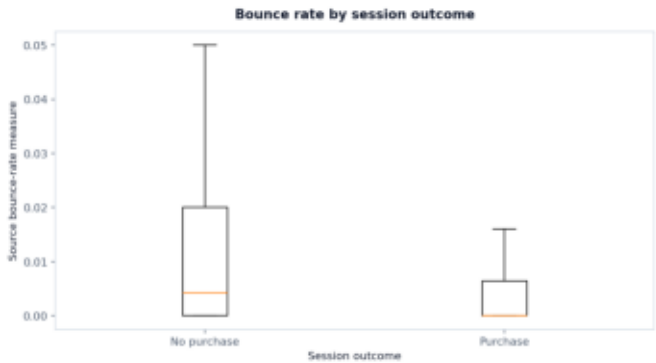
HEALTHCARE



SPORTS



FINANCE



E-COMMERCE

METHODS, SOURCES, AND REPRODUCTION

- Sales: eligible positive invoice lines, with cancelled invoices excluded; gross sales are quantity × price in GBP.
- Healthcare: encounter-level readmission within 30 days; expired discharge disposition excluded; patient-cluster bootstrap interval.
- Sports: non-copy team-game records; Elo forecast assessed on the final three archive seasons; RAPTOR ratings are source estimates.
- Finance: split-adjusted close price returns; 252-day annualization; complete-coverage cohort has selection/survivorship risk.
- E-commerce: session conversion with a Wilson interval and visitor-type association test; no repeat-purchase or basket inference.

Selected sources and terms

Online Retail II — UCI Machine Learning Repository; 10.24432/C5CG6D; CC BY 4.0

Diabetes 130-US Hospitals 1999-2008 — UCI Machine Learning Repository; 10.24432/C5230J; CC BY 4.0

NBA Elo historical games — FiveThirtyEight; master file snapshot retrieved 2026-11-04; CC BY 4.0

Historical RAPTOR by player — FiveThirtyEight; master file snapshot retrieved 2026-11-04; CC BY 4.0

New York Stock Exchange historical prices — Kaggle dataset author; source version not available; mirror master snapshot; CC0

Online Shoppers Purchasing Intention — UCI Machine Learning Repository; 10.24432/C5F88Q; CC BY 4.0

Reproduce with docs/setup.md and the documented domain builders. Full provenance and caveats: datasets/catalog.csv and docs/data_dictionary.md.